

Problems on Function

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python function that accepts a name and returns a greeting message.

Input	name = Asha
Expected Output	Hello, Asha!

2. Write a Python function that accepts two numbers and returns their sum.

Input	a = 12 b = 8
Expected Output	Sum = 20

3. Write a Python function that returns the greater of two numbers.

Input	a = 35 b = 48
Expected Output	Greater number = 48

4. Write a Python function that checks whether a number is even or odd.

Input	number = 26
Expected Output	Even

5. Write a Python function that returns the square of a number.

Input	number = 7
Expected Output	Square = 49

6. Write a Python function to calculate the area of a rectangle.

Input	length = 12 width = 5
Expected Output	Area = 60

7. Write a Python function to convert Celsius temperature to Fahrenheit.

Input	Celsius = 25
Expected Output	Fahrenheit = 77.0

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8. Write a Python function to count the number of vowels in a string.

Input	text = Programming
Expected Output	Vowel count = 3

9. Write a Python function to reverse a given string.

Input	text = Python
Expected Output	nohtyP

10. Write a Python function to calculate the factorial of a number.

Input	number = 5
Expected Output	Factorial = 120

11. Write a Python function to return the sum of all numbers in a list.

Input	numbers = [10, 20, 30, 40]
Expected Output	Sum = 100

12. Write a Python function to calculate the average of numbers in a list.

Input	numbers = [10, 20, 30, 40]
Expected Output	Average = 25.0

13. Write a Python function to check whether a string is a palindrome.

Input	text = madam
Expected Output	True

14. Write a Python function to calculate simple interest.

Input	principal = 10000 rate = 8 time = 2 years
Expected Output	Simple Interest = 1600.0

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15. Write a Python function with a default greeting message that can also accept a custom message.

Input	name = Ravi message not provided
Expected Output	Hello, Ravi!

16. Write a Python function to calculate the final price after applying a discount percentage.

Input	price = 2500 discount = 10%
Expected Output	Final price = 2250.0

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python function that returns the minimum, maximum, and average of a list in a single call.

Input	numbers = [12, 5, 30, 18, 9]
Expected Output	Minimum = 5 Maximum = 30 Average = 14.8

2. Write a Python function using *args to return the sum of any number of arguments.

Input	values = 10, 20, 30, 40, 50
Expected Output	Sum = 150

3. Write a Python function using **kwargs to format employee information.

Input	name='Asha', role='Developer', city='Pune'
Expected Output	Name: Asha, Role: Developer, City: Pune

4. Write a Python program with a function that updates and returns a global counter every time it is called.

Input	Call the function 3 times
Expected Output	Counter values = 1, 2, 3

5. Write a Python function to count the frequency of each element in a list and return a dictionary.

Input	items = ['A', 'B', 'A', 'C', 'B', 'A']
Expected Output	{'A': 3, 'B': 2, 'C': 1}

6. Write a recursive Python function to find the sum of digits of a number.

Input	number = 5832
Expected Output	Sum of digits = 18

7. Write a recursive Python function to return the nth Fibonacci number.

Input	n = 8 Sequence starts with 0, 1
Expected Output	Fibonacci number = 21

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8. Write a Python function to calculate income tax using slabs: 0% up to 300000, 5% for the next 300000, and 10% for the remaining amount.

Input	income = 800000
Expected Output	Tax = 35000.0

9. Write a Python function containing a nested helper function to calculate gross salary from basic salary, HRA, and DA percentages.

Input	basic = 30000 HRA = 20% DA = 10%
Expected Output	Gross salary = 39000.0

10. Write a Python function to validate a password and return both status and reason. Require at least 8 characters, one uppercase letter, and one digit.

Input	password = Python123
Expected Output	Valid = True Reason = Password accepted

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a recursive Python function to flatten a nested list containing values at different nesting levels.

Input	data = [1, [2, [3, 4], 5], [6, 7]]
Expected Output	[1, 2, 3, 4, 5, 6, 7]

2. Write a Python function to perform binary search on a sorted list and return the index of the target value.

Input	numbers = [2, 5, 8, 12, 16, 23, 38] target = 16
Expected Output	Index = 4

3. Write a Python function to merge overlapping intervals.

Input	intervals = [(1, 3), (2, 6), (8, 10), (9, 12)]
Expected Output	[(1, 6), (8, 12)]

4. Write a Python function to find the longest common prefix among a list of strings.

Input	words = ['flower', 'flow', 'flight']
Expected Output	Longest common prefix = fl

5. Write a Python function to group anagrams together.

Input	words = ['eat', 'tea', 'tan', 'ate', 'nat', 'bat']
Expected Output	[['eat', 'tea', 'ate'], ['tan', 'nat'], ['bat']] (group order may vary)

6. Write a Python function to return the k most frequent elements in a list.

Input	numbers = [1, 1, 1, 2, 2, 3] k = 2
Expected Output	[1, 2]

7. Write a Python function to generate the nth row of Pascal's Triangle.

Input	n = 5 First row is n = 1
Expected Output	[1, 4, 6, 4, 1]

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8. Write a Python function that behaves like a simple calculator based on an operation argument.

Input	a = 20 b = 5 operation = 'divide'
Expected Output	Result = 4.0

9. Write a Python function to multiply two matrices and return the resulting matrix.

Input	A = [[1, 2], [3, 4]] B = [[5, 6], [7, 8]]
Expected Output	[[19, 22], [43, 50]]

10. Write a Python function to validate whether brackets in an expression are balanced.

Input	expression = '{}[]'
Expected Output	Balanced = True